Agile Software Development

Lecture 01

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Software Development Life Cycle

- **Software Development Life Cycle** (SDLC) is a structured approach to the development of software.
- Several activities are done sequentially to achieve the end product.
- Each phase is associated with a deliverable that acts as an input to the subsequent phase of SDLC.

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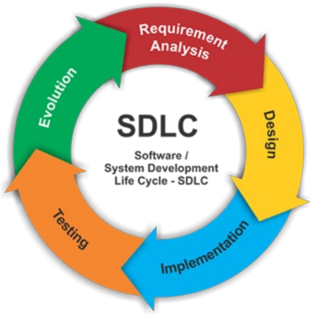
Phases of SDLC



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SDLC Models

- During the years of the SDLC evolution, different models were developed.
  - Waterfall model
  - Iterative model
  - Spiral model
  - V-shaped model
  - Agile model

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Waterfall Model

- Waterfall is a cascade SDLC model in which development process looks like the flow, moving step by step through the different phases.
- This model includes gradual execution of every stage completely.
- This process is strictly documented and predefined with features expected to every phase of this SDLC model.

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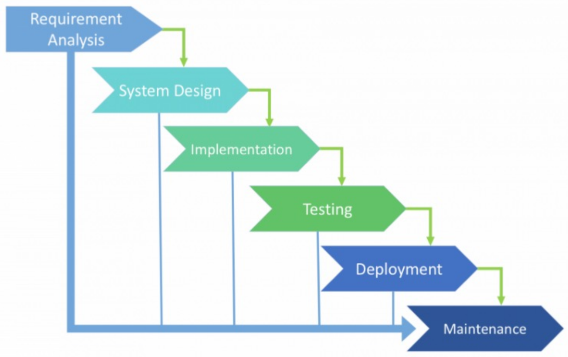
Waterfall Model



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Waterfall Model

- Use of waterfall model is suitable when;
  - the requirements are precisely documented.
  - the product definition is stable.
  - the technologies stack is predefined which makes it not dynamic.
  - the requirements are not ambiguous.
  - the project is short.

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Iterative Model

- This model does not need the full list of requirements before the project starts.
- Process is repetitive, allowing to make new versions of the product for every cycle.
- Every cycle includes the development of a separate component of the system.
- At the end of iteration this component is added to the functional developed earlier.

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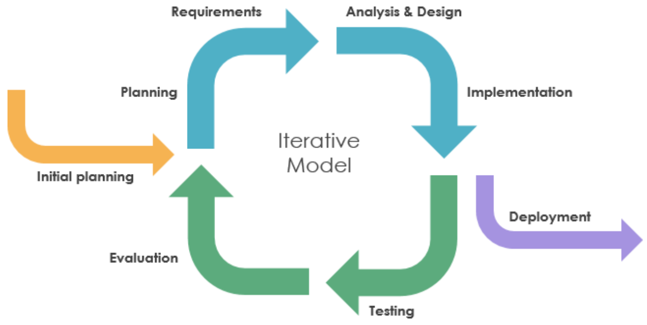
Iterative Model



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Iterative Model

- Use of Iterative model is suitable when;
  - the requirements to the final product are strictly predefined.
  - applied to the large-scale projects.
  - the main task is predefined, but the details may advance with the time.

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Spiral Model

- This model is a combination of the Iterative and waterfall SDLC models.
- Each cycle involves the same sequence of steps as the waterfall model.
- Spiral model has 4 quadrants.
  1. Determine objectives, alternatives and constraints.
  2. Evaluate alternatives, identify and resolve risks.
  3. Develop next-level product.
  4. Plan next phase.

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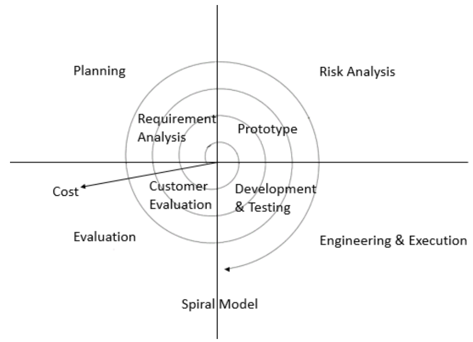
Spiral Model



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Spiral Model

- Use of Spiral model is suitable when;
  - the customer isn't sure about the requirements.
  - major changes are expected during the development cycle.
  - the projects is with mid or high-level risk.
  - the project is new product that should be released in a few stages to have enough of clients feedback.

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V-shaped Model

- This model is a model is an expansion of classic waterfall model.
- This is a very strict model and the next stage is started only after the previous phase.
- It based on associated test stage for the every development stage.
- Also called "Validation and Verification" model.

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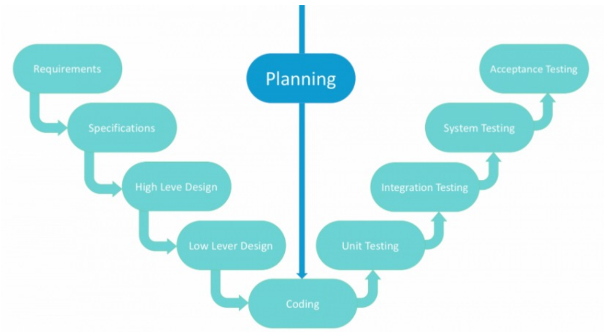
V-shaped Model



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V-shaped Model

- Use of V-shaped model is suitable;
  - for the projects where an accurate product testing is required.
  - for the small and mid-sized projects, where requirements are strictly predefined.
  - when the engineers of the required qualification, especially testers, are within easy reach.

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Agile Model

- Agile is set of **methods and methodologies** that help project team to think more effectively, work more efficiently, and make better decisions.
- Each of those methods and methodologies consists of **practices** that are streamlined and optimized to make them as easy as possible to adopt.

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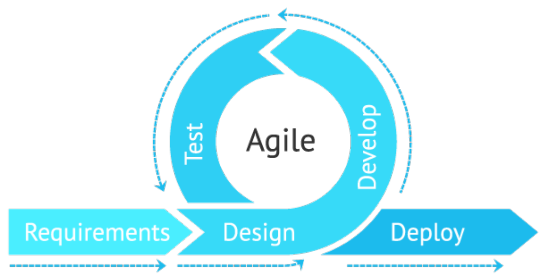
Agile Model



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Agile Model

- Use of Agile model is suitable;
  - the users' needs change dynamically.
  - Less price for the changes implemented because of the many iterations.
  - Unlike the Waterfall model, it requires only initial planning to start the project.

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An Agile Mindset

- Agile is also a **mindset**.
- A right mindset can make a big difference in how effectively a team uses the practices.
- This mindset helps people on a team share information with one another, so that they can make important project decisions together.

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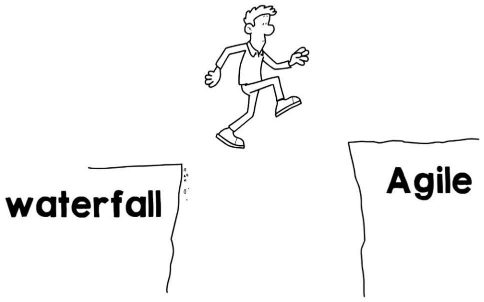
Going Agile ?



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An Agile Mindset

- For many teams that have not had as much success.
  - Achieved some improvements
  - Not substantial changes that they feel
- Key to that difference is often the mindset the team brings to each project.
- “Going agile” means helping the team find an effective mindset.

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An Agile Mindset

- An agile mindset is about opening up planning, design, and process improvement to the entire team.
- An agile team uses practices in a way where everyone shares the same information, and each person on the team has a say in how the practices are applied.

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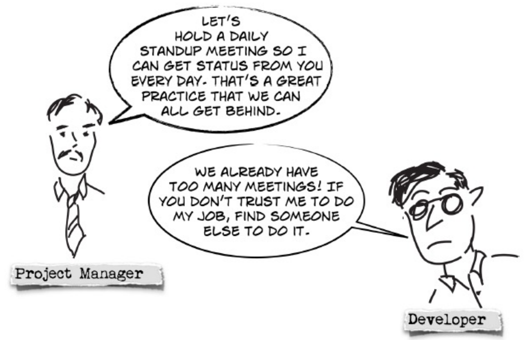
An Agile Mindset: Example



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An Agile Mindset: Example

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An Agile Mindset: Example

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An Agile Mindset: Example

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An Agile Mindset: Example

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An Agile Mindset: Example

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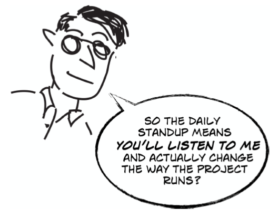
An Agile Mindset: Example

- When every person on the team feels like they have an equal hand in planning and running the project, the daily standup becomes more valuable—and much more effective.



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Case Study

Narrative: a team working on a streaming audio jukebox project

Dan – lead developer and architect

Bruce – team lead

Joanna – newly hired project manager

Tom – product owner

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Case Study

Activity

- Read the case study and list down the ineffective software practices that had been followed by Dan and Bruce in their last project.

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